IGCSE 0607 Practice Questions — E4.6 — Combined Questions & Markscheme

Perpendicular lines

Total Questions: 9 — **Total Marks:** 48 Marks

Table of Contents & Quick Navigation

- **E4.6** Perpendicular lines
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

1 Line L has the equation $5y + 2x = 15$.
Find the gradient of a line that is perpendicular to L.

..... **[2]**

2 Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L_1 has the equation $y = 5 - 2x$.
Line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$.

(a) Find the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 A is the point $(-4, 7)$ and B is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[5]**

5 A is the point $(-3, 8)$ and B is the point $(9, -2)$.

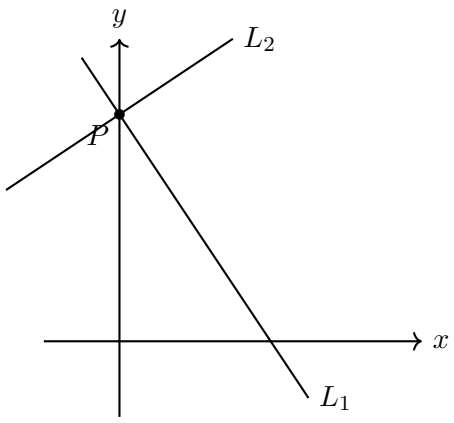
(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[4]**

6 Line L_1 has the equation $3x + 2y = 12$.
Line L_2 is perpendicular to L_1 and intersects L_1 at a point P on the y-axis.



Not to scale

(a) Find the coordinates of point P.

(..... , ...) **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Point Q lies on L_2 and has an x-coordinate of 9.
Point R lies on L_1 and has an x-coordinate of 4.
Work out the area of triangle PQR.

..... **[5]**

7 The equation of line P is $3x - 4y = 12$.
Line Q is perpendicular to line P and passes through the point $(6, -2)$.

Find the equation of line Q.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

8 The points A and B have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

9 The line L_1 has equation $2y + 3x = 12$.
The line L_2 is perpendicular to L_1 and intersects L_1 at the point P on the y-axis.

(a) Find the coordinates of P.

(..... , ...) **[2]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line L_2 crosses the x-axis at the point Q.
Calculate the exact length of PQ.

..... **[4]**

1 Rearranges to $5y = -2x + 15$ to find gradient of L is $-\frac{2}{5}$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\frac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\frac{1}{3}x + 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

3 (a) -2 **[B1]**

(b) Gradient of $L_2 = \frac{1}{2}$ **[M1]**

$1 = \frac{1}{2}(4) + c$ or $y - 1 = \frac{1}{2}(x - 4)$ **[M1]**

$y = \frac{1}{2}x - 1$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g., $y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

5 (a) $\left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(b) Gradient of $AB = \frac{-2-8}{9-(-3)}$ or $-\frac{10}{12}$ or $-\frac{5}{6}$ **[M1]**

Perpendicular gradient is $\frac{6}{5}$ **[M1]**

Substitute $(3,3)$ into $y-3=\frac{6}{5}(x-3)$ or $3=\frac{6}{5}(3)+c$ **[M1]**

$6x-5y=3$ or $-6x+5y=-3$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 (a) $(0,6)$ **[B1]**

(b) Gradient of $L_1=-\frac{3}{2}$ **[M1]**

Gradient of $L_2=\frac{2}{3}$ **[M1]**

$y=\frac{2}{3}x+6$ **[A1]**

(c) Coordinates of Q: $y=\frac{2}{3}(9)+6=12 \implies Q(9,12)$ **[B1]**

Coordinates of R: $3(4)+2y=12 \implies y=0 \implies R(4,0)$ **[B1]**

Identify triangle PQR is right-angled at P OR setup valid box/shoelace method **[M1]**

Length $PQ=\sqrt{9^2+6^2}=\sqrt{117}$ and Length $PR=\sqrt{4^2+(-6)^2}=\sqrt{52}$ **[M1]**

Area $=\frac{1}{2}\times\sqrt{117}\times\sqrt{52}=39$ **[A1]**

[Back to Question] *[Total Marks: 9]*

7 $4y=3x-12$ implies gradient of P is $\frac{3}{4}$ **[M1]**

Gradient of Q is $-\frac{4}{3}$ **[M1]**

Substitute $(6,-2)$ into $y=-\frac{4}{3}x+c$ or $y-(-2)=-\frac{4}{3}(x-6)$ **[M1]**

$-\frac{4}{3}x+6$ **[A1]**

[Back to Question] *[Total Marks: 4]*

8 (a) $\left(\frac{-2+6}{2},\frac{4+0}{2}\right)$ **[M1]**

$(2,2)$ **[A1]**

(b) Gradient of $AB=\frac{0-4}{6-(-2)}=-\frac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $=2$ **[M1]**

Substitute midpoint $(2,2)$ into $y-y_1=m(x-x_1)$: $y-2=2(x-2)$ **[M1]**

$y=2x-2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

9 (a) Substitutes $x=0$ into $2y+3x=12$ because P is on the y-axis **[M1]**

$(0, 6)$ **[A1]**

(b) Rearranges L_1 equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for L_2 is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

(c) Substitutes $y = 0$ into the equation of L_2 to find the coordinates of Q **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Back to Question] *[Total Marks: 9]*